A INTRODUCTION

A1 Classification Societies today

The purpose of a Classification Society is to provide classification and statutory services and assistance to the maritime industry and regulatory bodies as regards maritime safety and pollution prevention, based on the accumulation of maritime knowledge and technology.

The objective of ship classification is to verify the structural strength and integrity of essential parts of the ship's hull and its appendages, and the reliability and function of the propulsion and steering systems, power generation and those other features and auxiliary systems which have been built into the ship in order to maintain essential services on board. Classification Societies aim to achieve this objective through the development and application of their own Rules and by verifying compliance with international and/or national statutory regulations on behalf of flag Administrations.



The vast majority of commercial ships are built to and surveyed for compliance with the standards laid down by Classification Societies. These standards are issued by the Society as published Rules.

A vessel that has been designed and built to the appropriate Rules of a Society may apply for a certificate of classification from that Society.

However, such a certificate does not imply, and should not be construed as, a warranty of safety, fitness for purpose or seaworthiness of the ship. It is an attestation only that the vessel is in compliance with the Rules that have been developed and published by the Society issuing the classification certificate. Further, Classification Societies are not guarantors of safety of life or property at sea or the seaworthiness of a vessel because the Classification Society has no control over how a vessel is manned, operated and maintained between the periodical surveys which it conducts.

More than 50 organizations worldwide define their activities as providing some form of marine classification services; however, not all meet the definition given in Appendix 1. Some that do (listed in Appendix 2) form the International Association of Classification Societies (IACS). It is estimated that the Members of IACS collectively class over 90 percent of all commercial tonnage involved in international trade worldwide.

Classification is one element within the maritime safety regime. Others with a responsibility for or interest in promoting maritime safety include shipowners, shipbuilders, flag State administrations, port State control authorities, underwriters, shipping financiers, charterers, and, of course, seafarers.

The role of classification and Classification Societies has been recognized in the International Convention for the Safety of Life at Sea, (SOLAS) and in the 1988 Protocol to the International Convention on Load Lines. This statutory role is addressed later in this note.

As an independent, self-regulating, externally audited, body, a Classification Society has no commercial interests related to ship design, ship building, ship ownership, ship operation, ship management, ship maintenance or repairs, insurance, or chartering. In establishing its Rules, each Classification Society may draw upon the advice and review of members of the industry and academia who are considered to have relevant knowledge or experience.

Classification Rules are developed to establish standards for the structural strength of the ship's hull and its appendages, and the suitability of the propulsion and steering systems, power generation and those other features and auxiliary systems which have been built into the ship to assist in its operation. Classification Rules are not intended as a design code and in fact cannot be used as such.

A vessel built in accordance with the applicable Rules of an IACS Member Society may be assigned a class designation by the Society on satisfactory completion of the relevant surveys. For ships in service, the Society carries out surveys to verify that the ship remains in compliance with those Rules. Should any defects that may affect class become apparent, or damages be sustained between the relevant surveys, the owner is required to inform the Society concerned without delay.

The classification of a vessel is based on the understanding that the vessel is loaded, operated and maintained in a proper manner by competent and qualified crew or operating personnel.

A vessel may be maintained in class provided that, in the opinion of the Society concerned, it remains in compliance with the relevant Rules, as ascertained by periodic or non-periodic survey.

In developing its Rules, a Classification Society typically relies on empirical experience gained from classing a wide variety of ship types over many years, coupled with appropriate research that contributes towards the on-going development of relevant, advanced technical requirements.

Classification Societies are often simply referred to as 'Class Societies' or just 'Class' ('class').

A2 Why 'Classification'?

In the second half of the 18th century, marine insurers, based at Lloyd's coffee house in London, developed a system for the independent technical assessment of the ships presented to them for insurance cover. In 1760 a Committee was formed for this purpose, the earliest existing result of their initiative being *Lloyd's Register Book* for the years 1764-65-66.

At that time, an attempt was made to 'classify' the condition of each ship on an annual basis. The condition of the hull was classified A, E, I, O or U, according to the excellence of its construction and its adjudged continuing soundness (or otherwise). Equipment was G, M, or B: simply, good, middling or bad. In time, G, M and B were replaced by 1, 2 or 3, which is the origin of the well-known expression 'A1', meaning 'first or highest class'.

The concept of classification slowly spread to other countries and insurance markets. Bureau Veritas (BV) was founded in Antwerp in 1828, moving to Paris in 1832. 'Lloyd's Register of British and Foreign Shipping' was reconstituted as a self-standing 'Classification Society' in 1834; Rules for construction and survey were published the same year.

RINA (previously Registro Italiano Navale) dates from 1861; American Bureau of Shipping (ABS) traces its origins back to 1862. Adoption of common Rules for ship construction by Norwegian insurance societies in the late 1850s led to the establishment of Det Norske

Veritas (DNV) in 1864. Germanischer Lloyd (GL) was formed in 1867 and Nippon Kaiji Kyokai (ClassNK) in 1899. The Russian Maritime Register of Shipping (RS) was an early offshoot of the River Register of 1913. More recent foundations have been Polish Register of Shipping (PRS) in 1936; Yugoslav Register of Shipping (now the Croatian Register of Shipping (CRS)), 1949; China Classification Society (CCS), 1956; Korean Register of Shipping (KR), 1960; and Indian Register of Shipping (IRS), 1975.

As the classification profession evolved, the practice of assigning different classifications has been superseded, with some exceptions. Today a vessel either meets the relevant Class Society's Rules or it does not. As a consequence it is either 'in' or 'out' of 'class'. However, each of the Classification Societies has developed a series of notations that may be granted to a vessel to indicate that it is in compliance with some additional voluntary criteria that may be either specific to that vessel type or that are in excess of the standard classification requirements.

A3 The International Association of Classification Societies - IACS

IACS can trace its origins back to the International Load Line Convention of 1930 and its recommendations. The Convention recommended collaboration between Classification Societies to secure "*as much uniformity as possible in the application of the standards of strength upon which freeboard is based…*".

Following the Convention, RINA hosted the first conference of major Societies in 1939 - also attended by ABS, BV, DNV, GL, LR and NK - which agreed on further cooperation between the Societies.

A second major Class Society conference, held in 1955, led to the creation of Working Parties on specific topics and, in 1968, to the formation of IACS by seven leading Societies. The value of their combined level of technical knowledge and experience was quickly recognised. In 1969, IACS was given consultative status with the International Maritime Organization (IMO). It remains the only non-governmental organization with Observer status which is able to develop and apply Rules.

Compliance with the IACS Quality System Certification Scheme (QSCS) is mandatory for IACS Membership. Full details of the scheme are available on the IACS website.



IACS is governed by a Council, with each Member represented by a senior management figure.

Under the Council is the General Policy Group (GPG), made up of a senior manager from each Member, which develops and implements actions giving effect to the policies, directions and long term plans of the Council.

The chair of GPG is taken by the Member holding the Council chair. IACS's technical work is undertaken generally through specialist Working Groups overseen by GPG.

The Association maintains a Secretariat in London and a QSCS Operations Centre in Southampton, UK.

The IACS Charter, Procedures, details of the work programme, technical Resolutions and other publications are all available on the IACS website.